PHYSICS - CHAPTER 1

Electric Charges and Fields

1. Introduction to Electric Charge

Electric charge is a fundamental property of matter that causes it to experience a force when placed in an electromagnetic field. There are two types of charges: Positive and Negative.

- Like charges repel each other.
- Unlike charges attract each other.
- **Quantization of Charge:** Charge is always an integral multiple of the basic quantum of charge e. Formula:

$$q = ne$$

where $n = \pm 1, \pm 2, \ldots$ and $e \approx 1.602 \times 10^{-19}$ C.

Conservation of Charge

The total charge of an isolated system remains constant. Charge can neither be created nor destroyed; it can only be transferred from one body to another.

2. Coulomb's Law

Coulomb's law states that the electrostatic force between two stationary point charges is directly proportional to the product of their magnitudes and inversely proportional to the square of the distance between them.

$$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2}$$

Where ϵ_0 is the permittivity of free space ($\approx 8.854 \times 10^{-12}\ \mathrm{C^2N^{-1}m^{-2}}$).

Property	Electrostatic Force	Gravitational Force
Nature	Attractive or Repulsive	Always Attractive
Dependence on Medium	Depends on the intervening medium	Independent of the medium
Strength	Very Strong ($\sim 10^{36}$ times stronger)	Relatively Weak

3. Electric Field & Dipole

The electric field $\mathbf{E}$ at a point is defined as the electrostatic force $\mathbf{F}$ experienced by a small positive test charge q_0 placed at that point, divided by the charge itself.

$$\mathbf{E} = \lim_{q_0 \to 0} \frac{\mathbf{F}}{q_0}$$

Figure 1.1: Electric field lines of an electric dipole (a pair of equal and opposite charges).

Electric Field Lines Rule

Electric field lines **never** cross each other. If they did, it would mean the electric field has two different directions at the point of intersection, which is physically impossible.

4. Electric Flux and Gauss's Law

Electric flux Φ_E through a surface is the total number of electric field lines passing through that surface.

$$\Phi_E = \oint_S \mathbf{E} \cdot d\mathbf{A}$$

Gauss's Law

The total electric flux through any closed surface is equal to $1/\epsilon_0$ times the net charge enclosed by the surface.

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{encl}}}{\epsilon_0}$$

Important Applications of Gauss's Law

1. **Infinitely Long Straight Charged Wire:**

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

2. **Infinite Uniformly Charged Plane Sheet:**

$$E = \frac{\sigma}{2\epsilon_0}$$

3. **Uniformly Charged Thin Spherical Shell (Radius R):**
   Outside ($r > R$):

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

   Inside ($r < R$):

$$E = 0$$